

Chapter 14: Protection

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- Goals of Protection
- Principles of Protection
- Domain of Protection
- Access Matrix
- Implementation of Access Matrix
- Access Control

Goals of Protection

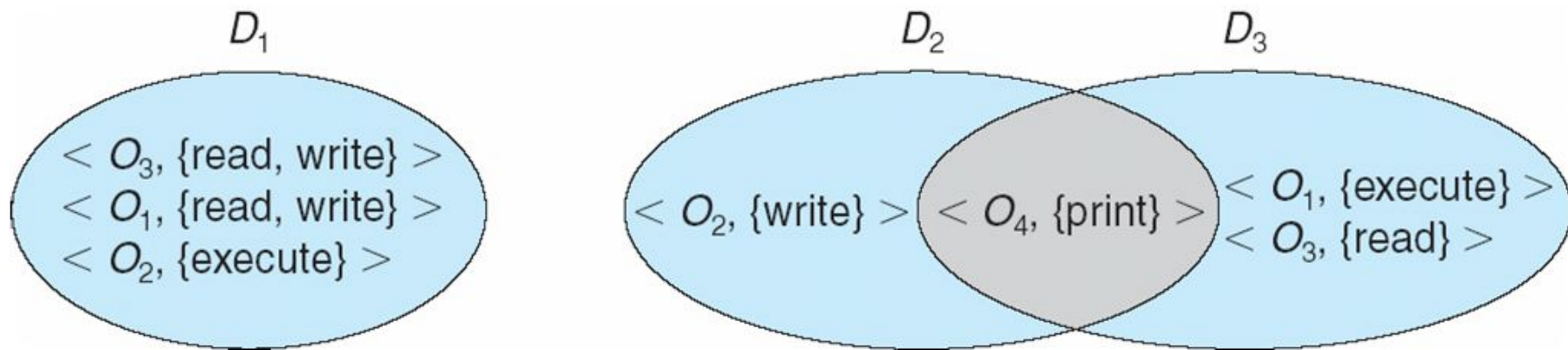
- Operating system consists of a collection of objects.
- Objects can be hardware objects(such as the CPU, memory segments, printers, disks, and tape drives), or software objects(such as files, programs, and semaphores).
- Each object has a unique name and can be accessed through a well-defined set of operations
- Protection problem - ensure that each object is accessed correctly and only by those processes that are allowed to do so
- A process should be allowed to access only those resources for which it has authorization. At any time, a process should be able to access only those resources that it currently requires to complete its task. This second requirement, commonly referred to as the **need-to-know principle**, is useful in limiting the amount of damage a faulty process can cause in the system

Principles of Protection

- Guiding principle – principle of least privilege
 - Programs, users and systems should be given **just enough** privileges to perform their tasks

Domain Structure

- A process operates within a **protection domain** which specifies the resources that the process may access.
- Each domain defines a set of objects and the types of operations that may be invoked on each object.
- The ability to execute an operation on an object is an **Access right**.
- Access-right = $\langle \text{object-name}, \text{rights-set} \rangle$
where *rights-set* is a subset of all valid operations that can be performed on the object.
- Domain = set of access-rights



Domain Implementation (UNIX)

- System consists of 2 domains:
 - User
 - Supervisor
- UNIX
 - Domain = user-id
 - Domain switch accomplished via file system
 - 4 Each file has associated with it a domain bit (setuid bit)
 - 4 When file is executed and setuid = on, then user-id is set to owner of the file being executed. When execution completes user-id is reset

Access Matrix

- View protection as a matrix (*access matrix*)
- Rows represent **domains**
- Columns represent **objects**
- $\text{Access}(i, j)$ is the set of operations that a process executing in Domain_i can invoke on Object_j

Access Matrix

domain \ object				
	F_1	F_2	F_3	printer
D_1	read		read	
D_2				print
D_3		read	execute	
D_4	read write		read write	

Use of Access Matrix

- If a process in Domain D_i tries to do “op” on object O_j , then “op” must be in the access matrix
- Can be expanded to dynamic protection
 - Operations to add, delete access rights
 - Special access rights:
 - 4 *owner of O_i*
 - 4 *copy op from O_i to O_j*
 - 4 *control – D_i can modify D_j access rights*
 - 4 *transfer – switch from domain D_i to D_j*

Use of Access Matrix (Cont)

- Access matrix design separates mechanism from policy
 - Mechanism
 - 4 Operating system provides access-matrix + rules
 - 4 If ensures that the matrix is only manipulated by authorized agents and that rules are strictly enforced
 - Policy
 - 4 User dictates policy
 - 4 Who can access what object and in what mode

Access Matrix of Figure A With Domains as Objects

domain \ object	F_1	F_2	F_3	laser printer	D_1	D_2	D_3	D_4
D_1	read		read			switch		
D_2				print			switch	switch
D_3		read	execute					
D_4	read write		read write		switch			

Figure B

Access Matrix with Copy Rights

object domain	F_1	F_2	F_3
D_1	execute		write*
D_2	execute	read*	execute
D_3	execute		

(a)

object domain	F_1	F_2	F_3
D_1	execute		write*
D_2	execute	read*	execute
D_3	execute	read	

(b)

Access Matrix With Owner Rights

object domain	F_1	F_2	F_3
D_1	owner execute		write
D_2		read* owner	read* owner write
D_3	execute		

(a)

object domain	F_1	F_2	F_3
D_1	owner execute		write
D_2		owner read* write*	read* owner write
D_3		write	write

(b)

Modified Access Matrix of Figure B

object domain	F_1	F_2	F_3	laser printer	D_1	D_2	D_3	D_4
D_1	read		read			switch		
D_2				print			switch	switch control
D_3		read	execute					
D_4	write		write		switch			